

AI Image & Voice Generation Using Open-Source Tools

Generative AI Internship – Week 1

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Part A: AI Poster Generation

Task 1: Prompt Engineering

Education

A futuristic AI-powered classroom with holographic displays, students learning through augmented reality, ultra realistic, cinematic lighting, 4K.



Healthcare

Advanced AI hospital with robotic doctors assisting patients, modern medical technology, 3D render, high detail.



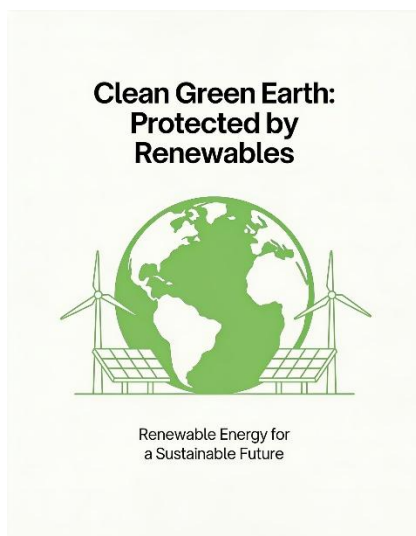
Agriculture

Smart farming using drones and AI crop monitoring, green fields, realistic photography, professional poster.



Environment

Clean green earth protected by renewable energy, minimalist poster design, eco-friendly theme.



Artificial Intelligence

Futuristic AI city with glowing neural networks, cyberpunk style, neon lighting, ultra detailed.

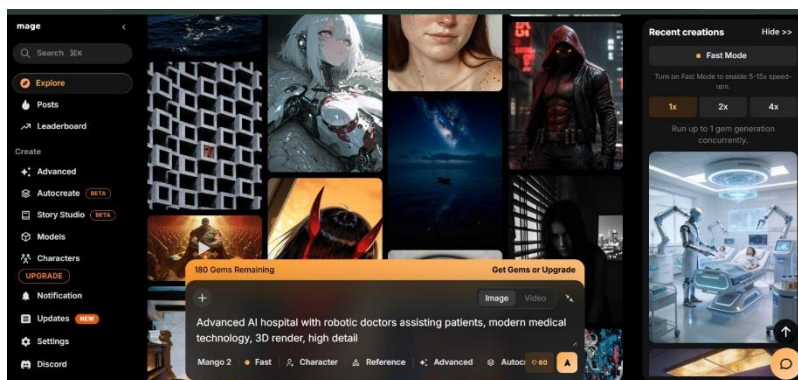


Task 2: Generated Posters

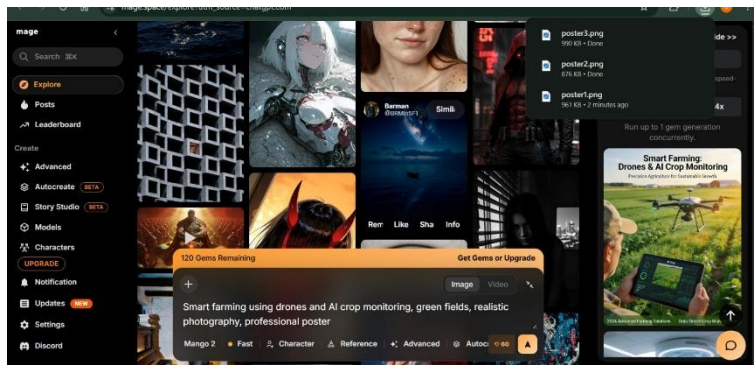
Poster 1 - Education



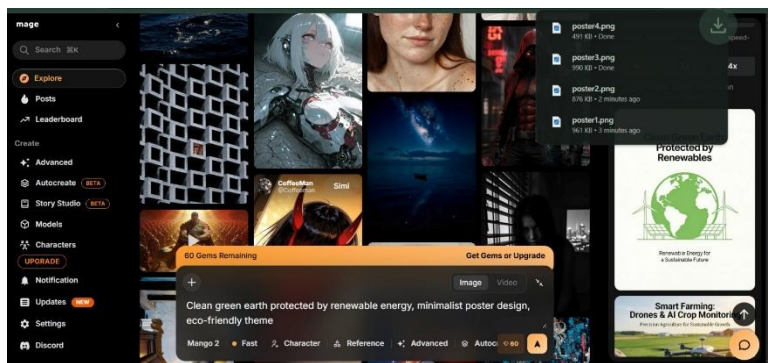
Poster 2 - Healthcare



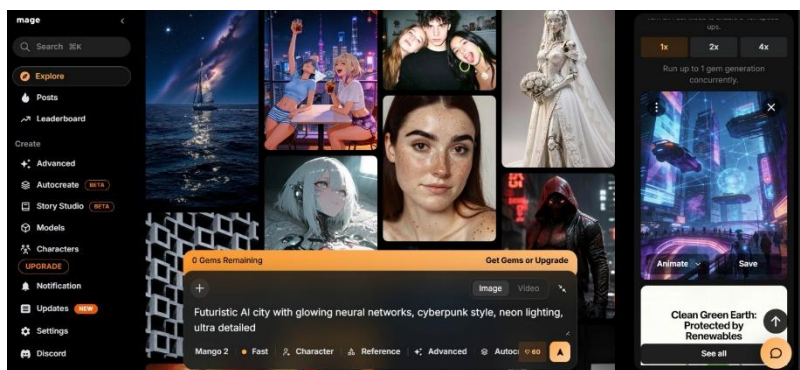
Poster 3 - Agriculture



Poster 4 - Environment



Poster 5 - Artificial Intelligence



Task 3: Poster Analysis

Poster	Style	Output Quality	Improvements
Education	Realistic	High quality classroom visualization	Add more interactive AI devices
Healthcare	3D Render	Detailed hospital environment	Improve character realism

Poster	Style	Output Quality	Improvements
Agriculture	Realistic	Clear smart farming representation	Add more autonomous machines
Environment	Minimalist	Clean and attractive design	Increase color contrast
Artificial Intelligence	Cyberpunk	Highly detailed futuristic city	Add more AI visual elements

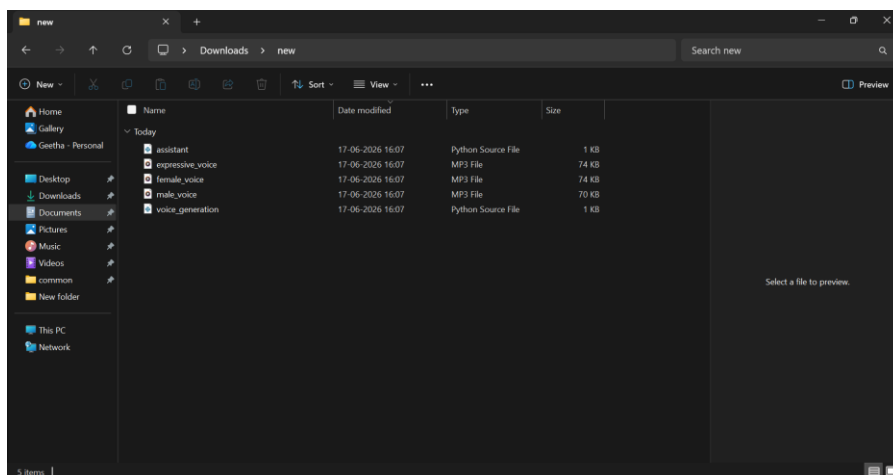
Part B: AI Voice Generation

Task 1: Text-to-Speech

Generated speech for:

"Welcome to the Generative AI Internship Program. Today we explore AI-powered image and voice generation technologies."

Evidence

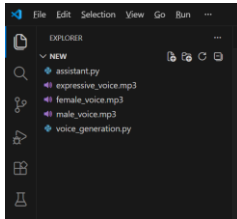


Task 2: Multi-Voice Generation

Generated:

- Male Voice
- Female Voice
- Expressive Voice

Evidence



Task 3: Voice Analysis

Model	Voice Type	Quality	Pros	Cons
GTTS Default	Male	Good	Clear pronunciation	Limited emotion
GTTS UK	Female	Good	Natural accent	Similar tone
GTTS Australia	Expressive	Very Good	Pleasant voice	Limited customization

Part C: AI Voice Assistant

Objective

Developed a simple voice assistant that:

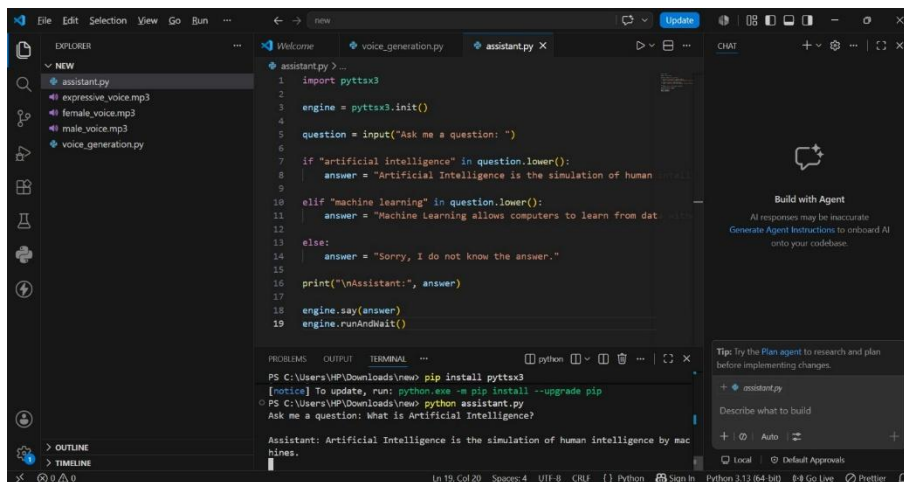
- Accepts text input
- Generates a response
- Converts response into speech
- Plays generated audio

Example

User: What is Artificial Intelligence?

Assistant: Artificial Intelligence is the simulation of human intelligence by machines.

Evidence



```
1 import pyttsx3
2
3 engine = pyttsx3.init()
4
5 question = input("Ask me a question: ")
6
7 if "artificial intelligence" in question.lower():
8     answer = "Artificial Intelligence is the simulation of human intelligence by machines."
9
10 elif "machine learning" in question.lower():
11     answer = "Machine Learning allows computers to learn from data and make predictions."
12
13 else:
14     answer = "Sorry, I do not know the answer."
15
16 print("\nAssistant:", answer)
17
18 engine.say(answer)
19 engine.runAndWait()
```

PROBLEMS OUTPUT TERMINAL

PS C:\Users\HP\Downloads\new> pip install pyttsx3
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\HP\Downloads\new> python assistant.py
Ask me a question: What is Artificial Intelligence?
Assistant: Artificial Intelligence is the simulation of human intelligence by machines.

Chat

Build with Agent
AI responses may be inaccurate.
Generate Agent Instructions to onboard AI onto your codebase.

Tip: Try the Plan agent to research and plan before implementing changes.

+ assistant.py
Describe what to build
+ | Auto | +

Part D: Research Component

Comparison of Stable Diffusion, DALL·E and Midjourney

Parameter	Stable Diffusion	DALL·E	Midjourney
Cost	Free	Païd	Païd
Open Source	Yes	No	No
Image Quality	Excellent	Excellent	Excellent
Customization	Very High	Medium	Low
Hardware Requirements	High	Low (Cloud)	Low (Cloud)

Research Conclusion

Stable Diffusion, DALL·E, and Midjourney are powerful AI image generation systems. Stable Diffusion is open-source and provides extensive customization options, making it popular among developers and researchers.

DALL·E offers high-quality image generation through cloud-based services and is easy to use. Midjourney is known for producing visually appealing artistic outputs with minimal prompt engineering. Stable Diffusion requires stronger hardware resources but offers maximum flexibility. DALL·E and Midjourney provide convenience through cloud infrastructure. Each tool has its own advantages depending on user requirements and technical expertise.

Conclusion

This assignment provided practical experience in AI-powered image generation, voice synthesis, and voice assistant development. Stable Diffusion demonstrated how text prompts can be transformed into visually appealing posters. gTTS enabled the creation of realistic speech outputs from text. A simple voice assistant was developed to answer user queries and generate spoken responses. The project highlighted the capabilities of open-source generative AI tools and their applications in marketing, automation, content creation, and intelligent assistant systems. Overall, the assignment improved understanding of generative AI technologies and their real-world applications.

References

- [Stable Diffusion](#)
- [gTTS Documentation](#)
- [Python Documentation](#)
- [Open Source Generative AI Resources](#)